MIL-P-46633A (MU)
30 November 1962
SUPERSEDING
MIL-P-46633 (Ord)
5 March 1962

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, M470 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of projectile designated as Projectile, HE, 155MM, M470.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-R-11470 - Radiographic Inspection Qualification of Equipment, Operators and Procedures.
MIL-R-11471 - Radiographic Inspection of Metals.
MIL-I-45208 - Inspection Requirements, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

FSC: 1320

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DRAWINGS

ORDNANCE CORPS

8861068 - Projectile, 155MM, HE, M470, Loading Assembly.
8837839 - Pallet for Projectile, 155MM, HE, M470.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL - 8861068 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.

3.1.1 General.-Material and parts shall be in accordance with applicable specifications and drawings.

3.1.2 Clean scrap.-Clean scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.2).

3.2 Loaded projectile.-The loaded projectile shall comply with all requirements specified on Drawing (dwg.) 8861068, and with all requirements specified in applicable specifications.

3.3 High explosive charge.

3.3.1 Cavities and cracks.

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3.3.1.1 Cracks and projected cavities.-Cavities having a projected length of 1/32 inch (in.) or less shall be disregarded, except no base separations shall be permitted between the base of the shell cavity and the charge. Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.3.1.

TABLE I

	Segments (See Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.).	*1/64	1/4	1/2	1/2
Projected length of any cavity, excluding pipes, cracks and annular rings (in.).	1/8	1/2	1/2	3/4
Piping cavities max. projected area (sq. in.).	0	1/4	1/2	0
Piping cavities, max. projected width (in.).	0	1/4	1/4	0
Cracks, max. projected width (in.).	1/32	1/32	1/32	1/32
Annular rings, max. projected width (in.)	0	0	1/4	0

* If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch.

3.3.1.2 Surface cavitation.-Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The max. dimension of any visible cavity in the faced surface of the charge shall not exceed 1/4 inch

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in any direction and the total area of all cavities shall not exceed 80 percent of the surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/32 inch or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by dwg. 8861068).

3.3.2 Porous areas.-Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.3 Cracked charges.-Not more than four transverse cracks shall be permitted in any charge, and not more than one transverse crack shall be permitted in Segment A.

3.4 Loading of high explosive.

3.4.1 Mating surfaces.-The mating surfaces of threads shall be free of caked explosives, and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care should be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.4.2 Solid explosives.-Solid explosives shall not be present in the projectile before loading or be added to the shell during pouring or cooling.

3.5 Coating.-Projectile having the interior wall imperfectly coated shall not be loaded. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile.

3.6 Workmanship.-The loaded projectile shall be free of chips, dirt, grease, rust and other foreign material and shall be painted as required by the projectile loading drawing. Care shall be exercised at all times so that the rotating band of each projectile will not be damaged by being nicked or burred.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of projectile loaded and assembled by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

a. Projectile metal parts from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

b. Loaded projectile of one weight zone.

c. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

d. Explosive charge of one interfix lot number from one manufacturer.

e. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Projectile loading assembly, prior to loading (see dwg. 8861068).

Categories	Defects	Method of Inspection	Code No. (See 6.5)
Critical:			
1.	Foreign matter in charge cavity (see 3.4.2)	Visual	01001
2.	Unsolidified pool of paint in bottom of cavity	Visual-Manual	01002
Major:	AQL 0.25 percent		
101.	Interior protective coating with bare spot (see 3.5)	Visual	01003
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	01004

4.2.2.2 Projectile loading assembly, prior to assembly of liner (see dwg. 8861068).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Depth of fuze well	Gage	02001
102.	Foreign matter in fuze well	Visual	02002
Minor:	AQL 1.50 percent		
201.	Diameter of fuze well, max	Gage	02003
202.	Surface cavitation excessive (see 3.3.1.2)	Visual	02004
203.	Disc missing	Visual	02005
204.	Evidence of poor workmanship (see 3.6)	Visual	02006

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4.2.2.3 Projectile loading assembly prior to assembly of lifting plug (see dwg. 8861068)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Supplementary charge pad missing	Visual	03001
Major:	AQL 0.40 percent		
101.	Concentricity of inside diameter of liner with pitch diameter of thread including perpendicularity with face of body	Gage	03002
102.	Depth to bottom of liner, minimum	Gage	03003
103.	Supplementary charge missing or inverted	Visual	03004
104.	Supplementary charge not removable by hand	Manual	03005
Minor:	AQL 4.0 percent		
201.	Spacer missing or damaged	Visual	03006
202.	Liner missing or damaged	Visual	03007
203.	Supplementary charge damaged	Visual	03008
204.	Explosive in threads (see 3.4.1)	Visual	03009
205.	Foreign matter in liner cavity including explosive	Visual	03010
206.	Liner removable by hand	Manual	03011
207.	Silicone compound missing from threads	Visual	03012
208.	Evidence of poor workmanship (see 3.6)	Visual	03013

4.2.2.4 Projectile loading assembly (see dwg. 8861068).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Projectile zone weight incorrect	Balance	04001
2.	Zone stenciling incorrect	Visual	04002

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Major:	AQL 0.40 percent		
101.	Torque test failure of lifting plug	Test	04003
102.	Diameter of bourrelet, max. (if retouched)	Gage	04004
103.	Lifting plug gasket missing	Visual	04005
104.	Any weight zone punch mark missing or inadequate	Visual	04006
Minor:	AQL 2.50 percent		
201.	Grommet missing, damaged or improperly assembled	Visual	04007
202.	Rotating band damaged (if grommet is missing)	Visual	04008
203.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	04009
204.	Marking misleading or unidentifiable (except for zones), including incorrect color code	Visual	04010
205.	Evidence of poor workmanship (see 3.6)	Visual	04011

4.2.2.5 Pallet (see dwg. 8837839).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Pallet damaged	Visual	05001
102.	DOD symbol missing or unidentifiable	Visual	05002
103.	Projectile of different zones on the same pallet	Visual	05003
104.	Strapping missing or loose	Visual-Manual	05004

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Minor:	AQL 1.50 percent		
201.	Number of projectile assemblies incorrect	Visual	05005
202.	Marking misleading or unidentifiable	Visual	05006
203.	Strapping improperly engaged	Visual	05007

4.2.3 Testing.

4.2.3.1 Cavitation (see 3.3 except 3.3.1.2) Critical defect-Code No. 02007.-The sampling plan and procedure for this test shall be in accordance with Handbook ORD-M608-11, using a CSP-1 plan having an AQL of 0.015 percent. Appropriate f, i and k values shall be as specified by the procuring activity. Any projectile having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.3.1, using equipment in accordance with 4.2.4.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Lists Number IEL 8861068 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3 Test methods and procedures.

4.3.1 Determination of cavities.-The sample projectile shall be subjected to x-ray examination. The x-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with Specification MIL-R-11470. In addition, the sensitivity of x-ray technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the projectile shall be re-x-rayed at an angle 90 degrees from that of the original exposure.

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4.3.1.1 Check of sensitivity of x-ray technique.-The sensitivity of x-ray technique shall be determined, by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the hole in the center, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thicknesses and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Specification MIL-R-11471. The actual thicknesses may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in Figure 2 of this specification.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards of radiographic densities be used (see 6.4). Comparison of film with standards may be made visually; however the use of a densitometer is recommended.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The projectile shall be packed in accordance with dwg. 8837839.

5.2 Marking.-Marking shall be in accordance with dwg. 8837839.

5.3 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification

b. Subject to the rights in data acquired under the contract, engineering drawings and lists provided and revised under this specification may be used by the Department of Defense activities for design, procurements, manufacture, testing evaluation, production, receiving and production inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever engineering drawings are needed.

6.2 Clean scrap.-Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment List in accordance with the instruction of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND.

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6.3.1.1 IEL abreviations.-The abreviations appearing on the IEL have the following significance:

DD - Detailed drawings
SCD - Specification control drawings
CI - Commercial items
CD - Contractor design

6.3.2 Detailed drawings.-Detailed drawings are drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. Contractors need provide no design when a detailed drawing is listed for an item of inspection equipment and the contractor elects to use the Government design. The contractor may, however, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND.

6.3.3 Specification control drawings.-Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND.

6.3.4 Commercial items.-Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

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6.3.5 Use of standard measuring equipment.-If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND, for approval.

6.4 Film density.-"American Standard For Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N.Y. has been found satisfactory as a source of information on film densities.

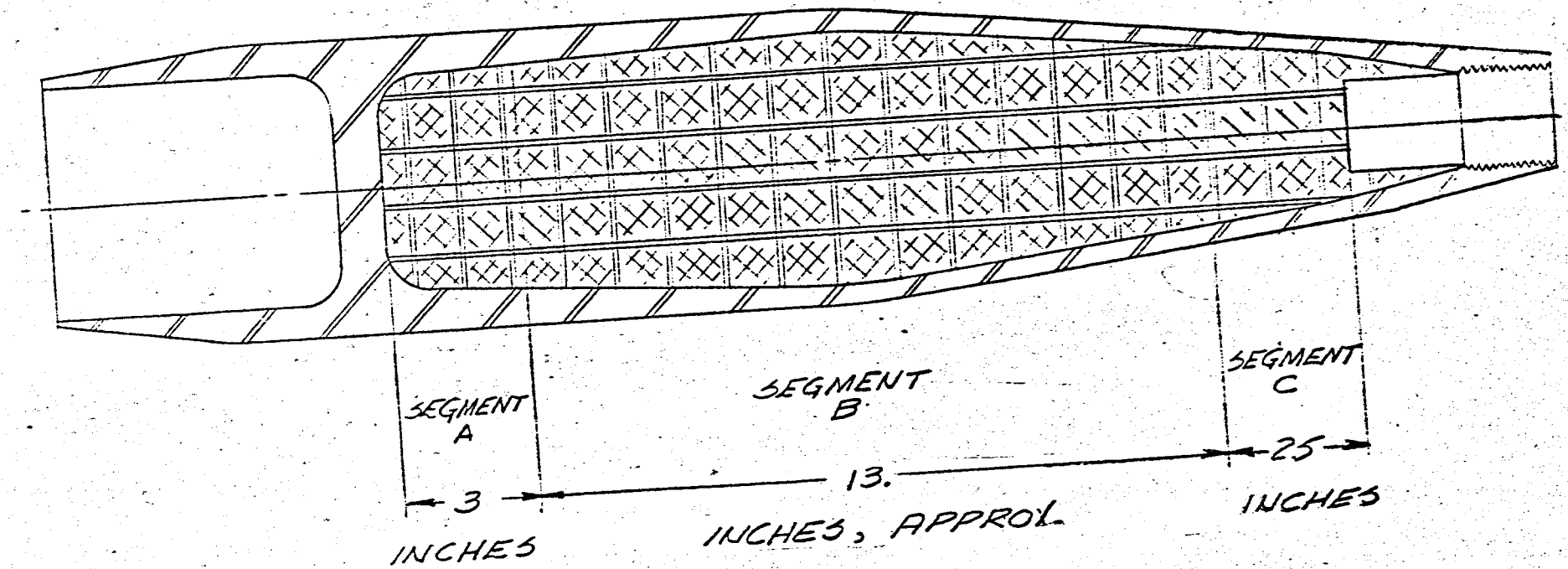
6.5 Inspection code numbers.-The five digit code numbers assigned to inspections herein are to facilitate future data collection and analysis by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-MU

Preparing activity:
Army-MU

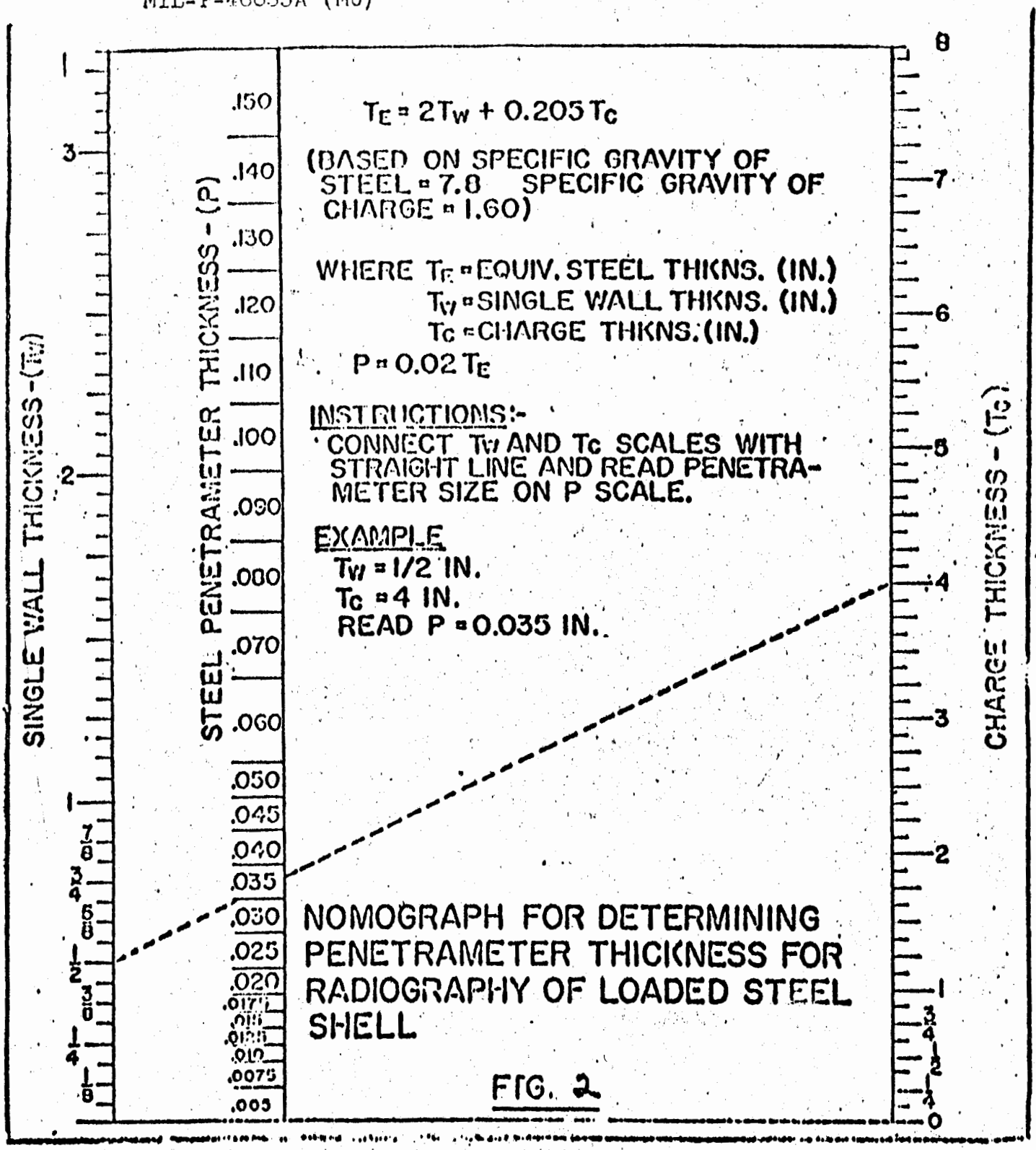
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SEGMENT D (FOR CHARGES CONTAINING A DEEP FUZE WELL ONLY) SHALL BE DEFINED AS EXTENDING FROM THE FORWARD BOUNDARY OF SEGMENT C, i.e. BASE OF DEEP FUZE WELL, TO A POINT 3.0 INCHES FROM THE NOSE OF THE PROJECTILE BODY.

FIGURE 1

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NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 2

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